

Application Layer Protocols (HTTP.SMTP/POP)

Examination Lab

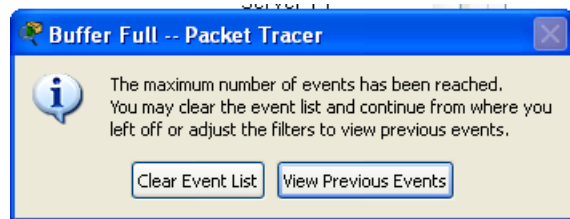
Objectives:

Capture traffic and observe the PDUS for HTTP, SMTP, POP.

Task 1: Observe HTTP traffic exchange between a client and server.

Step 1 – Run the simulation and capture the traffic.

- Enter **Simulation** mode.
- Click on the PC1. Open the **Web Browser** from the **Desktop**.
- Enter **www.bracu.ac.bd** into the browser. Clicking on **Go** will initiate a web server request. Minimize the Web Client configuration window.
- Two packets appear in the **Event List**, a DNS request needed to resolve the URL to the IP address of the web server and an ARP request needed to resolve the IP address of the server to its hardware MAC address.
- Click the **Auto Capture / Play** button to run the simulation and capture events.
- Sit tight and observe the packets flowing through the network.



- When the above message appears Click “View Previous Events”.
- Click on PC1. The web browser displays a web page appears.

Step 2 – Examine the following captured traffic.

Our objective in this lab is only to observe HTTP traffic.

	Last Device	At Device	Type
1.	PC1	Switch 0	HTTP
2..	Local Web Server	Switch 1	HTTP

- Find the following packets given in the table above in the **Event List**, and click on the colored square in the **Info** column.

(sec)	Last Device	At Device	Type	Info
--	PC1	Switch0	DNS	Red square
--	PC1	Switch0	ARP	Green square
PC1	Switch0	PC0	ARP	Green square
Switch0	Switch1	PC0	ARP	Green square
Switch0	Switch1	Switch1	ARP	Green square

- When you click on the Info square for a packet in the event list the **PDU Information** window opens. If you click on these layers, the algorithm used by the device (in this case, the PC) is displayed. View what is going on at each layer.

- Examine the PDU information for the remaining events in the exchange.

For packet 1::

What kind of HTTP packet is packet no. 1?

Packet no. 1 is an **HTTP GET request packet**. It is sent from the client (PC1) to the web server server to request the webpage www.bracu.ac.bd. The GET request asks the server to send the requested webpage to the client

Click onto “Inbound PDU details” tab. Scroll down at the end, what do you see?

At the end of the **Inbound PDU Details**, we can see the **HTTP GET request message** which contains the request line, HTTP version, and host information. This indicates that the client is requesting the webpage from the server.

For packet 2:

Click onto “Inbound PDU details” tab. Scroll down at the end, what do you see? What kind of HTTP packet is this?

At the end of the inbound PDU details we see the **HTTP response message** from the server..

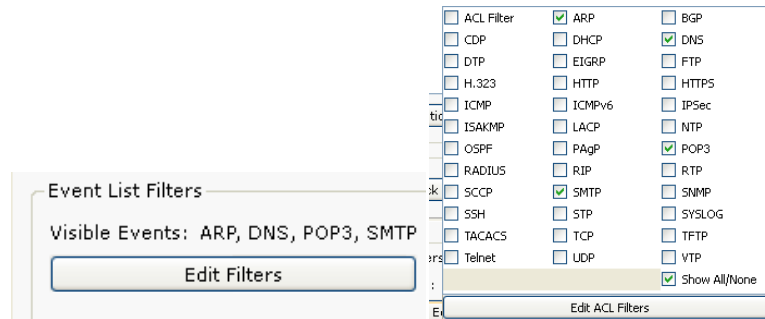
This packet is an **HTTP Response packet**. It usually contains a response status such as **HTTP/1.1. 200 OK**, which means the request was successful and the webpage is being sent to the client.

Task 2: Observe email traffic exchange between a client and email server using SMTP and POP3.

Step 1 – Run the simulation and capture the traffic.

- On the Event List window click “Reset Simulation” button. All previous packets will disappear.

- At the bottom of the Event List window, there is a filter which filters the protocols that we want to see. Click Edit filters. Another window appears showing different protocols, unclick HTTP and click SMTP and POP3.



- Click a space anywhere outside the popup window, then it will disappear.
- Your Event List Filter should be as shown below:

Event List Filters

Visible Events: ARP, DNS, POP3, SMTP

- Now click on the PC1. Close the web browser window. Open the **Email** from the **Desktop**. A mail browser window will open. Click “compose”, another window appears.

Compose Mail

To:

Subject:

- Fill the window as shown and press send.
- Minimize the client window .
- Click the **Auto Capture / Play** button to run the simulation and capture events.
- Sit tight and observe the packets flowing through the network.
- This interaction is between the sender client and its email server.

Step 2 – Examine the following captured traffic.

Our objective in this lab is only to observe SMTP traffic.

	Last Device	At Device	Type
3.	PC1	Switch 0	DNS
4.	PC1	Switch 0	SMTP
5.	Bracu Email Server	Switch 1	SMTP

- Find the following packets given in the table above in the **Event List**, and click on the colored square in the **Info** column.
- Examine the PDU information.

For packet 4::

What is the purpose of this DNS packet?

The purpose of this DNS packet is to **resolve the domain name of the email server into its IP address**. The client must know the IP address of the email server before sending the email using SMTP.

For packet 5& 6::

Explain why SMTP packet was sent to the email server and the server replied with an SMTP packet?

The **SMTP packet is sent by the client to the email server to deliver the email message.**

SMTP (Simple Mail Transfer Protocol) is responsible for sending emails. The email server then
replies with an SMTP response packet to confirm that the email has been received and processed
successfully.

Step 3 – Run the simulation and capture the traffic for POP.

- On the Event List window click “Reset Simulation” button. All previous packets will disappear.
- Now click on the PC0. Open the **Email** from the **Desktop**. A mail browser window will open. Click “**receive**”, minimize the window.
- Click the **Auto Capture / Play** button to run the simulation and capture events.
- Sit tight and observe the packets flowing through the network.
- This interaction is between the sender client and its email server.

Step 2 – Examine the following captured traffic.

Our objective in this lab is only to observe POP traffic.

	Last Device	At Device	Type
6.	PC1	Switch 0	DNS
7.	PC1	Switch 0	POP3
8.	Bracu Email Server	Switch 1	POP3

- Find the following packets given in the table above in the **Event List**, and click on the colored square in the **Info** column.
- Examine the PDU information.

For packet 6::

What is the purpose of this DNS packet?

The purpose of this DNS packet is to **find the IP address of the email server** so that the client can connect to it and retrieve emails using POP3.

For packet 7&8::

Explain why POP packet was sent to the email server and the server replied with a POP packet?

The **POP3** packet is sent by the client to request emails from the email server. POP3 (Post Office Protocol version 3) is used for receiving emails. The server replies with a POP packet containing the stored email messages and confirmation of the request.

